# Reflect with us - Week 7

This assignment will not be graded and is only for practice.

*1 point*

Let $L_1$ and $L_2$ be affine subspaces of $R^3$, where $L_1 = U$ and $L_2 = (2,0,1) + W$, for some vector subspaces $U$ and $W$ of $R^3$. Let a basis for $U$ be given by $\{(2,0,1),(1,1,0),(0,1,0)\}$ and a basis for $W$ be given by $\{(1,0,1),(0,1,1)\}$. Suppose there is a linear transformation $T: U \rightarrow W$ such that $(0,1,0) \in ker(T)$, $T(2,0,1) = (0,1,1)$ and $T(1,1,0) = (1,0,1)$. An affine mapping $f: L_1 \rightarrow L_2$ is obtained by defining $f(u) = (2,0,1) + T(u)$, for all $u \in U$. Which of the following options are true?

- [ ] $L_1 = R^3$
- [ ] $L_2 = \{(x,y,z) \mid x + y - z = 1\}$
- [ ] $L_2 = R^3$
- [ ] $f(x,y,z) = (x - 2z + 2, z, x - z + 1)$
- [ ] $f(x,y,z) = (x - 2z, z, x - z)$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

$L_1 = R^3$

$L_2 = \{(x,y,z) \mid x + y - z = 1\}$

$f(x,y,z) = (x - 2z + 2, z, x - z + 1)$

**Solution:**

Discussion on Option 1: It is given that $L_1 = U$ and $\{(2,0,1),(1,1,0),(0,1,0)\}$ is a basis of the vector subspace $U$.

The dimension of $U$ is

Hint

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**

(Type: Numeric) 3

*1 point*

$L_1 = U$ is a subspace of $R^3$. To know whether $L_1 = U$ is a proper subspac

R3 or not, we just have to compare their dimensions.

*1 point*

Now, can you see why $L_1 = U = R^3$ L1 = U = R3 ?

- ◯ Yes
- ◯ No

**No, the answer is incorrect.**
**Score: 0**
**Feedback:**

**Accepted Answers:**
Yes

Discussion on Options 2 and 3: Discussion on Options 2 and 3: It is given that $L_2 = (2,0,1) + W$ L2 = (2, 0, 1) + W and $\{(1,0,1),(0,1,1)\}$ {(1, 0, 1), (0, 1, 1)} is a basis for the vector subspace $W$ W. Now to find the explicit description of $L_2$ L2, we have to find the explicit description of $W$ W first.

$$\begin{aligned} W &= \{a(1,0,1) + b(0,1,1) \mid a, b \in R\} \\ &= \{(a, b, a+b) \mid a, b \in R\} \end{aligned}$$ W = {a(1, 0, 1) + b(0, 1, 1) | a, b ∈ R} = {(a, b, a + b) | a, b ∈ R}

As $W$ W is a subspace of $R^3$ R3, we want to express it using the usual $x, y, z$ x, y, z co-ordinate and the relation between them. From the above description we have, $x = a$ x = a, $y = b$ y = b, $z = a + b$ z = a + b, which implies that $z = x + y$ z = x + y. So

$$W = \{(x, y, z) \mid x + y - z = 0. \text{ where } x, y, z \in R\}$$ W = {(x, y, z) | x + y − z = 0. where x, y, z ∈ R}

*1 point*

Now, are you able to find the equation of $L_2$ L2?

- ◯ Yes
- ◯ No

**No, the answer is incorrect.**
**Score: 0**
**Feedback:**

**Accepted Answers:**
Yes

*1 point*

Do you agree that the sets $\{(2+x, y, z+1) | x + y - z = 0\}$ {(2 + x, y, z + 1) | x + y − z = 0} and $\{(a, b, c) \mid a + b - c = 1\}$ {(a, b, c) | a + b − c = 1} are the same?

- ◯ Yes
- ◯ No

**No, the answer is incorrect.**
**Score: 0**
**Feedback:**

**Accepted Answers:**
Yes

Discussion on Option 4 and 5:Discussion on Option 4 and 5:

Step 1:Step 1: Here, we have to find an algebraic expression for the affine map $f: L_1 \rightarrow L_2$. To get an algebraic expression for $f$, we have to find the algebraic expression which represents the linear transformation $T: U \rightarrow W$.

Given:

$$T(0,1,0) = (0,0,0), \quad T(2,0,1) = (0,1,1). \quad T(1,1,0) = (1,0,1)$$

*1 point*

Is $T$ uniquely determined by the above information?

◯ Yes

◯ No

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

**Accepted Answers:**

Yes

Step 2:Step 2: Since $\{(0,1,0),(2,0,1),(1,1,0)\}$ is a basis of $U = R^3$, we can write any element $(x,y,z) \in U$ as a linear combination of $\{(0,1,0),(2,0,1),(1,1,0)\}$.

$$(x,y,z)=a\, (0,1,0)+b\,(2,0,1)+c\,(1,1,0) \qquad (1)$$

Apply $T$ on both sides:

$$T(x,y,z) = aT(0,1,0) + bT(2,0,1) + cT(1,1,0)$$

$$T(x,y,z) = a(0,0,0) + b(0,1,1) + c(1,0,1)$$

To get an algebraic expression for $T(x,y,z)$, we just have to know the values of $a$, $b$ and $c$ in terms of $x$, $y$ and $z$.

*1 point*

Now, can you find the values of $a$, $b$ and $c$ using equation $(1)$?

◯ Yes

◯ No

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

**Accepted Answers:**

Yes

Solve the above system for $a, b$ and $c$.

*1 point*

What is the value of $a$ ?

◯

$z$

○ $x - 2z$

○ $y + 2z - x$

○ $x - y$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$y + 2z - x$

*1 point*
What is the value of $b$ ?

○ $z$

○ $x - 2z$

○ $y + 2z - x$

○ $x - y$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$z$

*1 point*
What is the value of $c$ ?

○ $z$

○ $x - 2z$

○ $y + 2z - x$

○ $x - y$

**No, the answer is incorrect.**
**Score: 0**
**Accepted Answers:**
$x - 2z$

Step 3:

$$T(x, y, z) = (0, z, z) + (x - 2z, 0, x - 2z) = (x - 2z, z, x - z)$$

By definition:

$$f(x, y, z) = (2, 0, 1) + T(x, y, z) \quad \text{where} (x, y, z) \in U.$$

By substituting the value $T(x, y, z)$ in the above expression, we get

$$f(x, y, z) = (x - 2z + 2, z, x - z + 1) \quad \text{for all}(x, y, z) \in U.$$
f(x, y, z) = (x − 2z + 2, z, x − z + 1) for all (x, y, z) ∈ U.

Check Answers